



FM50 – Centrifugal Pump Demonstration Unit

A motor-driven centrifugal pump, mounted on a stainless steel plinth with a water reservoir and pipework for continuous circulation.

PRODUCT CODE: FM50

Categories: Civil, Educational, FM Series - Fans And Compressors, FM51 - Series And Parallel Pumps Demonstration

Tags: Armfield , Centrifugal Pump Demonstration Unit , Continuous Circulation , FM50 , Impeller , Pump Volute , Single Stage , Water Reservoir

DESCRIPTION

A motor-driven centrifugal pump, mounted on a stainless steel plinth with a water reservoir and pipework for continuous circulation.

The pump volute and the water reservoir are manufactured from clear acrylic for maximum visibility. Similarly the pipe runs are made from transparent PVC. Manually operated valves at the pump inlet and outlet allow control of the flow and also facilitate the study of suction effects.

The pump volute has been designed so that the impeller can be easily accessed and replaced without tools.

The FM50 is delivered with two impellers, one with forward-curved blades and one with backward-curved blades, allowing the students to investigate the effects of impeller characteristics.

TECHNICAL SPECIFICATIONS

FEATURES & BENEFITS

Max flow rate: 1.6 l/s typical

Max head: 9.0m

Max pump speed: 1,800rpm

Motor power rating: 250W

Pressure sensor: 0 to 15psi

Flow sensor: 0 to 27l/s

Demonstration Capabilities

- Demonstration of a single-stage centrifugal water pump in operation
- Measurement of constant-speed pump performance, including production of characteristic curves:
 - Pump total head
 - Motor shaft power
 - Impeller speed
 - Pump efficiency
- Introduction to pump speed laws
- Investigation of impeller styles
- Comparison of student calculations with computer results

Ordering Specification

- A small-scale multi-stage centrifugal compressor demonstration unit, comprising an inlet duct, the compressor, an outlet duct and an adjustable aperture, all mounted on a stainless steel base
- Seven stages in the compressor
- Equipped with electronic measurement sensors for head pressure, flow rate (via orifice plate) and air temperature
- Capable of being linked to a PC (not supplied) via a USB interface console (an essential accessory), which does not require internal access to the computer. Also enables interfacing to other software packages
- Supplied with software providing full instructions for setting up, operating, calibrating and performing the teaching exercises. Facilities for logging, processing and displaying data graphically
- Offers a complete teaching package of coursework and laboratory investigation

Essential Accessories

Armfield IFD7

Requirements

- Armfield IFD7
- Software requires a computer running Windows XP or above with a USB port (computer not supplied by Armfield)

Shipping Specifications

Volume: 1.10m³

Gross Weight: 110Kg

Dimensions

Length: 0.88m

Width: 0.51m

Height: 0.60m

Order Codes

- FM50



- IFD7-A: 220-240V / 1Ph / 50Hz
- IFD7-G: 220-240V / 1Ph / 60Hz

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